39. THE LOCOMOTOR SYSTEM : DEVELOPMENT OF THE SOMITES

DURATION : 09:06 STUDY SHEET

COURSE SUMMARY

This video thoroughly explores the development of somites within the context of musculoskeletal system formation, emphasising the importance of the notochord, neural tube, and lateral plates. It addresses fundamental concepts such as the creation of the dermatome, myotome, and sclerotome, as well as their respective roles in the development of skin, muscles, and bone structure. The processes of molecular induction and epigenetics are also explained, highlighting how genetic and environmental factors interact to regulate this complex development.

DEVELOPMENT OF SOMITES IN THE MUSCULOSKELETAL SYSTEM

Initially, we studied the establishment of the **notochord**, then the **ectoderm**, with its associated **mesoderm**. The importance of this supporting tissue, which here is the intermediate tissue, was also discussed.

Next, we examined the **vitelline system**, which corresponds to our entire digestive system. We also began to study the **lateral plates**, including the **peritoneum**, **somatopleure**, and **splanchnopleure**.

The lateral plate, which ultimately forms the **renal**, **adrenal**, and **genital systems**, is a somatic plate. It is induced by the **epiblastic**, **chordal**, and **neural tube** levels.

This induction will cause a reorganisation of the somatic plate in several forms:

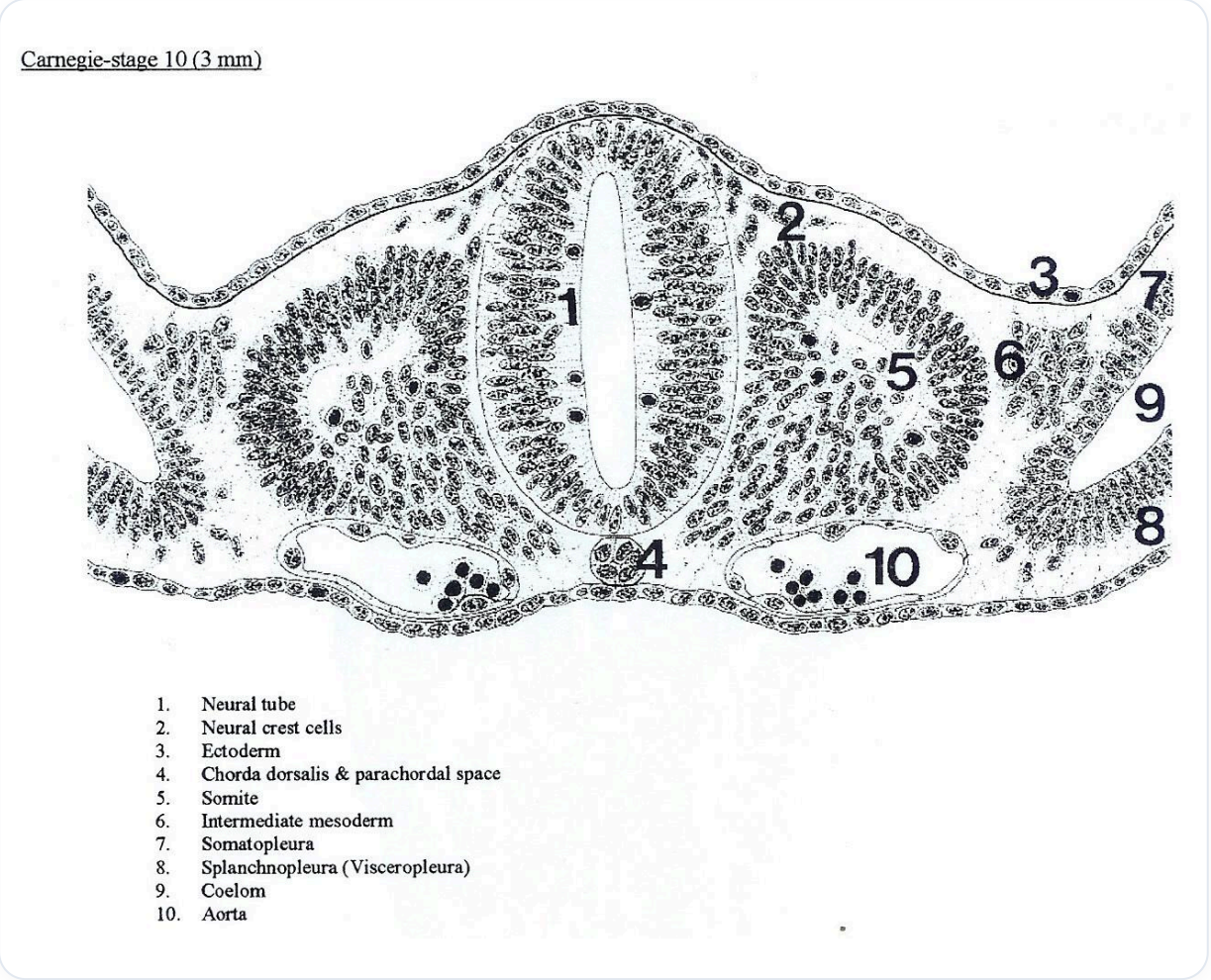


FIG. — Carnegie stage 10 embryo

- **Dermatome**: everything concerning the skin.
- **Myotome**: all muscles.
- **Sclerotome**: the vertebral axis.

The **fasciae** are found in the intermediate system, i.e., the peripheral tissue. They will be integrated by the myotome, which is muscular.

It is important to understand that there is **connective tissue** and **metabolic fields**. These fields are areas of dilation and retention, and the interaction between these systems will reorganise the whole.

It is a balance between the establishment of **bone** and **muscle**. Between the two, another field appears: a retention field, of a fascial nature.

The notochord influences the establishment of the neural tube, and this influence of the neural tube organises and reorganises the entire mesodermal structure, i.e., the somitic structure on either side of the neural tube.

We can therefore say that the notochord creates this **differential growth rate** within the peripheral and central parts of the neural plate, which will form a **groove**.

In this grooving process, a specific reorganisation of cells will occur. They will organise microscopically, with **induction factors**, to form the different parts of the somites.

The movement of the **aortas** influences the exorotation movement of the kidneys. At the same time, this movement will also play a role in the establishment of this somitic system.

The neural tube is made up of small cells. Do not forget that a tissue is made of cells, and cells are made of molecules.

So we have this whole combination. When we talk about **genetic factors**, we are at the molecular level. Cells will release new types of **proteins**, in the form of genetic **molecular enzymes**, such as **FGF8** or **sonic hedgehog (HH)**.

These proteins will provide the inductions, create the **ligands** and the necessary information to reorganise the system. The force comes from outside and will cause the reaction inside.

Genes are therefore not the engine, but the support for development. It is an **epigenetic** force.

As the aortas approach each other, through this differential growth rate and this movement of approximation, a kind of opening is observed, a movement where the cells, previously encapsulated, reorganise.

One part will form at the periphery through induction phenomena. Do not forget that there is an epiblast here. Molecular phenomena will occur, causing this part of the somite to become the **dermatome**.

The part located near the notochord will form the **sclerotome**. The sclerotome will form, among other things, the **vertebral disc**, and then the entire vertebra will take shape up to the spinous process, organised by this sclerotome coming from the somite.

Another intermediate part will follow another path: this will be the **myotome**, from which all muscles will derive.

So we have the **dermatome**, the **myotome**, and the **sclerotome**.

In this process of somite establishment, there is also a phenomenon of **segmentation**. Here, somites begin to appear. It is not just a tube; it becomes like small balls on either side.

This occurs when the embryo begins to lengthen, i.e., the neural tube develops. It is the neural tube that will induce.

But do not forget that at this moment, a phase sets in with the integration of the entire **vascular system**. Among other things, there was this system of **polarisation** where the ectodermal system, for the neural tube, was already in an assimilation field, beginning to create spaces.

It is therefore the vascular system that is at the origin of segmentation and not the neuronal system, as is often described. First, the neuronal system is needed, which will organise the vascular system, and from this neuronal system, the venous system will emerge.

Primary segmentation appears, as **Blechschmidt** very well described, through a region that develops as space and fragmentation of the system to form the somites. The embryo will have areas of **dermalgia**, and this is a segmentation: the somites segment.

It is "metamerised", if you will, it comes from the segmentation organised by the vascular system on a primitive plane, by the flexion and reabsorption of the aorta.

It is arterial, but in fact, at that level, these are **venous sinuses**. We will mainly say it is vascular, because initially, it is not yet truly defined as arterial or venous. It is a field of assimilation and dissimilation.

Do not forget that this **longitudinal growth** is allowed by aspiration, so it is polarity that performs it. Then, by enlarging, it will form what is called the **ectodermal ring**.

The embryo is like this. We are only just starting to have two or three first segmentations. And it is only when the flexion movement accelerates that blood will begin to enter the spaces that are being created.

Each level will begin to create what are called **somites**. There will be 33 pairs that will develop here.

This is an absorption by the vascular system, linked to a polarity between the ectoderm and the mesoderm. It is therefore polarity that ensures that each tissue, each space, must be nourished separately.

I cannot bring a vessel that is like this. No, it starts to happen like that. It is the vessels that create the space between the myotome and the sclerotome. It is the bone that has grown and pulls on the myotome. After that, it is the muscle.

So I have an inversion. Initially, the bone is fluid and grows. When it pulls and lengthens, it creates a metabolic field around it, which is a field of dilation. It is the bone that will become active first. And then, it will only be the muscle that will be active in a second phase.

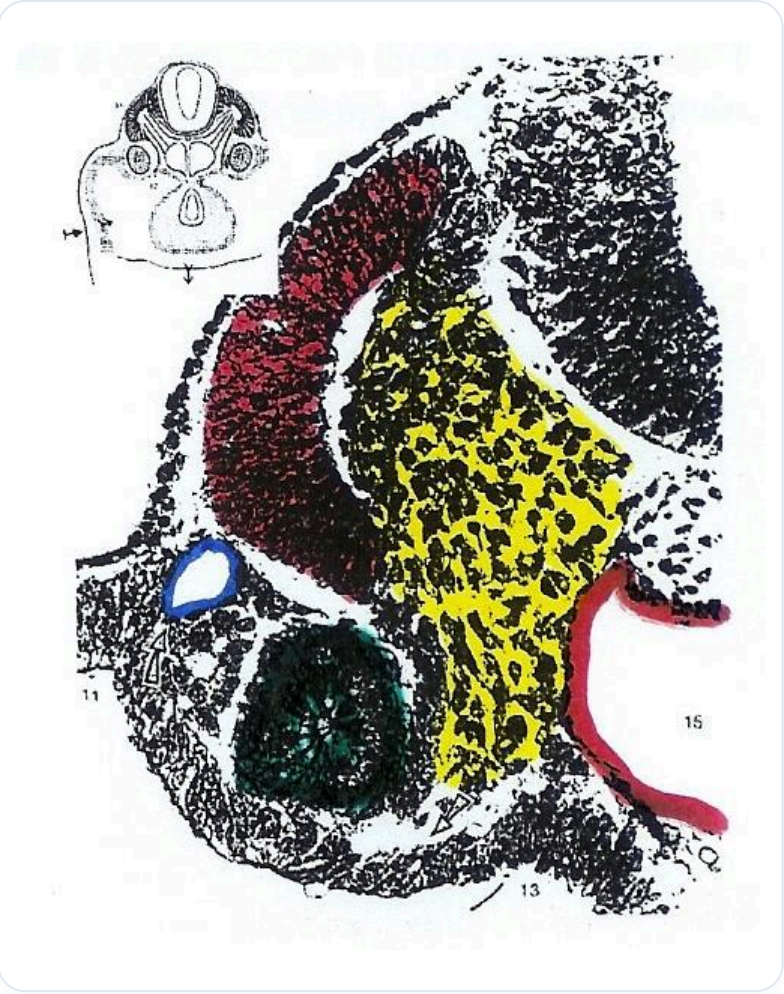


FIG. — Embryo: cross-section

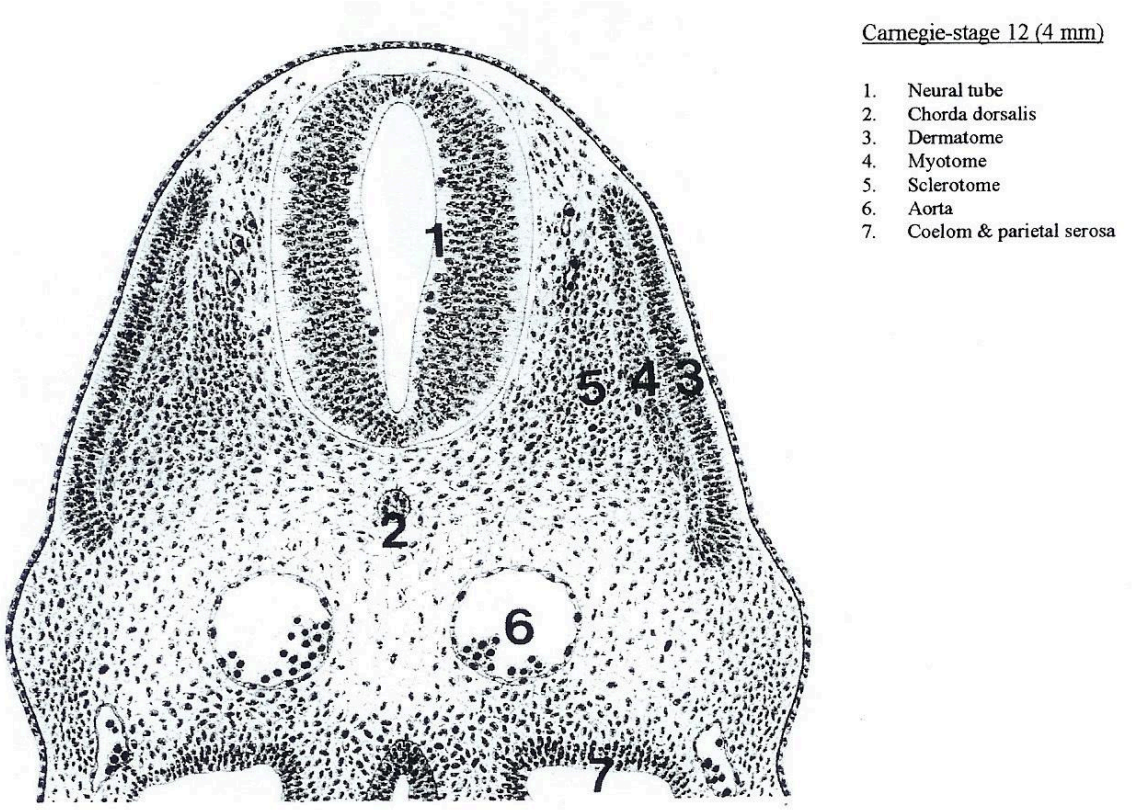


FIG. — Human embryo, cross-section, Carnegie stage 12

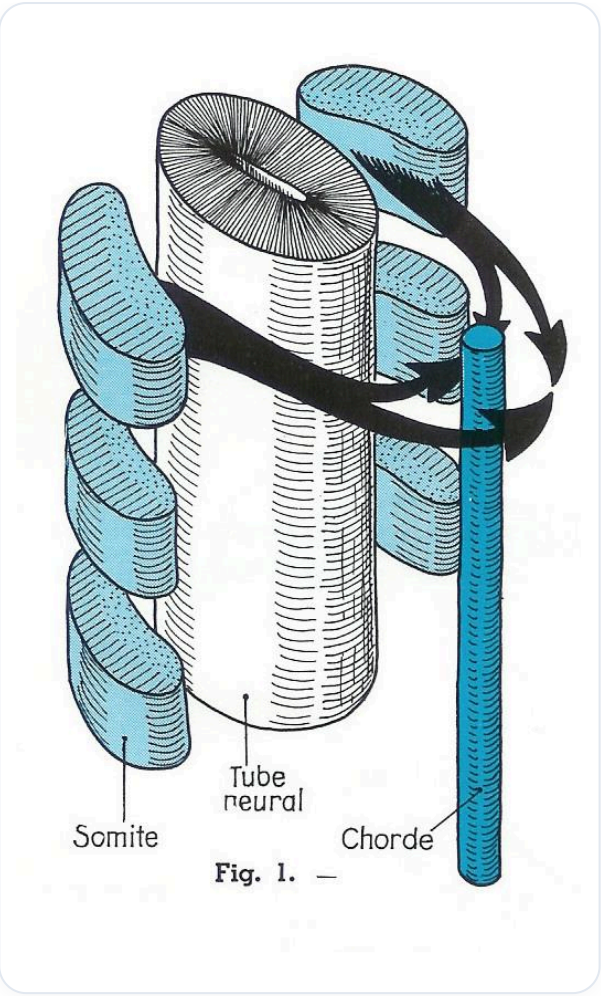


Fig. 1. —

FIG. — Neurulation and somites

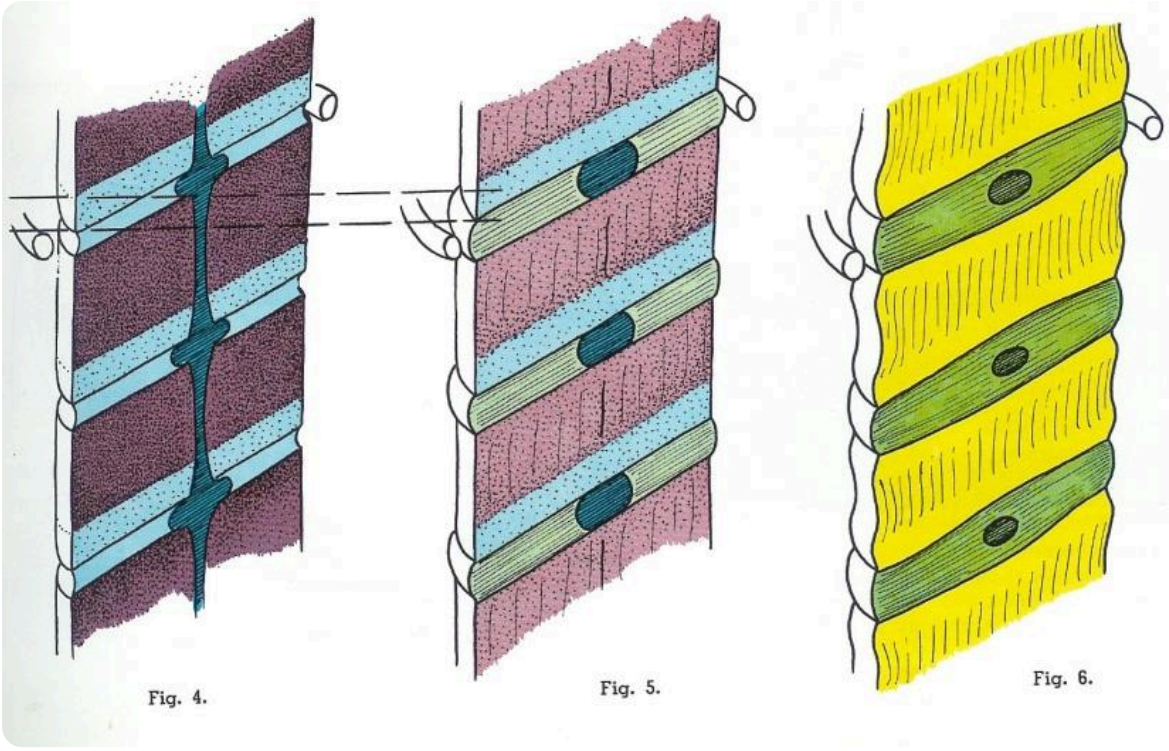


FIG. — Somite myotome dermatome

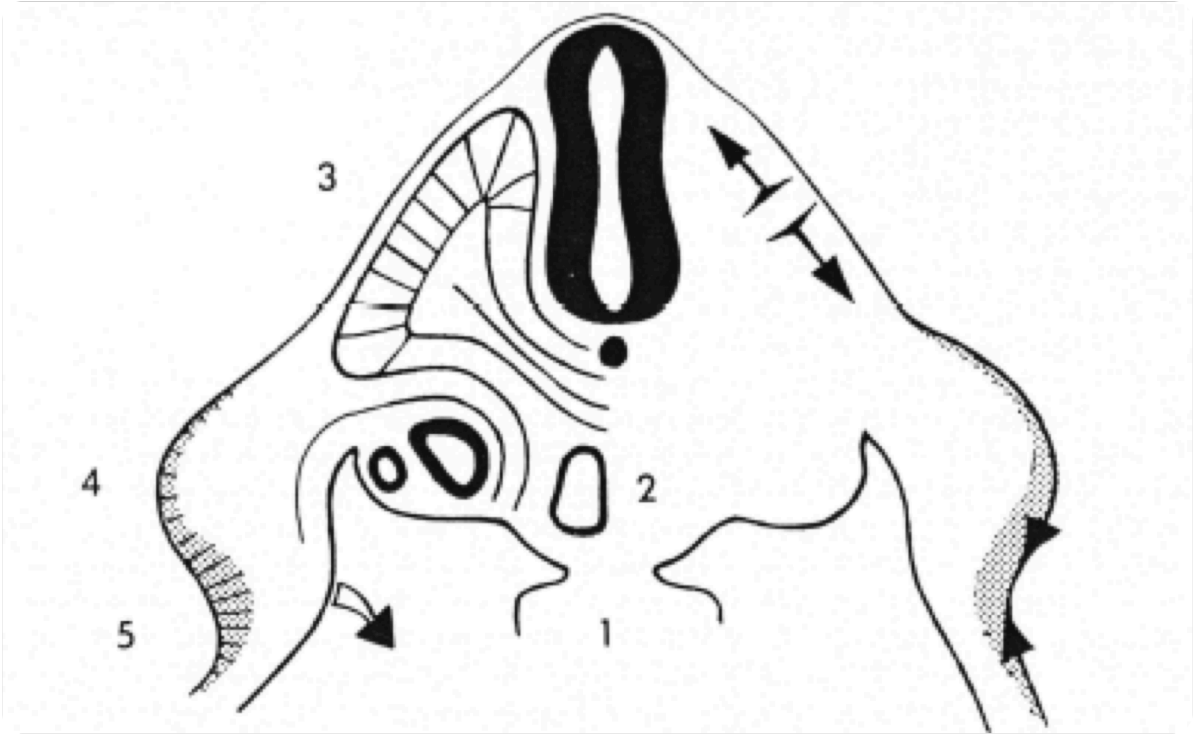


FIG. — Embryonic limb development

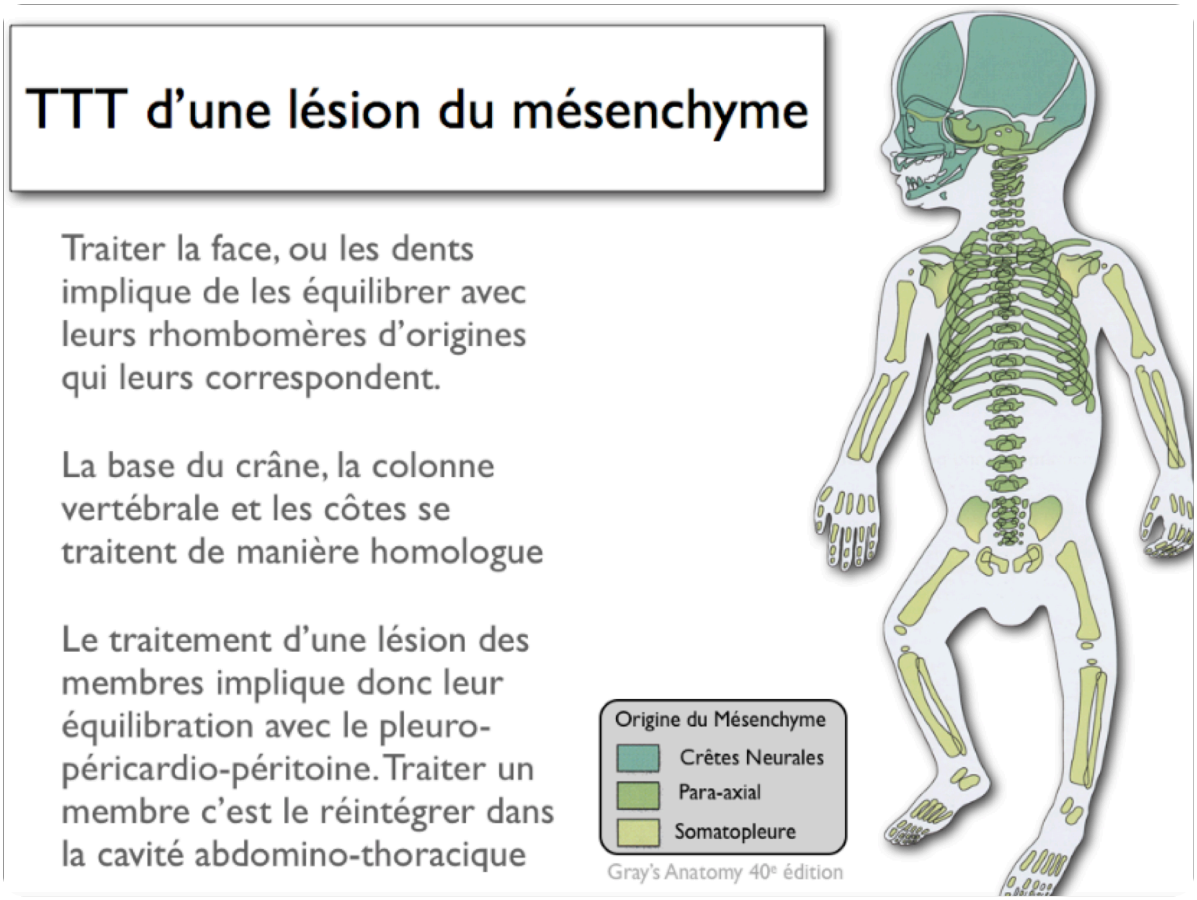


FIG. — Skeletal mesenchyme origin